

CHAPTER 5

PRIORITY QUEUES (HEAPS)

—— delete the element with the highest priority

§ 1 ADT Model

Objects: A finite ordered list with zero or more elements.

Operations:

☞ `PriorityQueue Initialize( int MaxElements );`

☞ `void  Insert( ElementType X, PriorityQueue H );`

☞ `ElementType  DeleteMin( PriorityQueue H );`

☞ `ElementType FindMin( PriorityQueue H );`

§ 2 Simple Implementations

Array :

Insertion — add one item at the end $\sim \Theta(1)$

Deletion — find the largest \ smallest key $\sim \Theta(n)$
remove the item and shift array $\sim O(n)$

Linked List :

Insertion — add to the front of the chain $\sim \Theta(1)$

Deletion — find the largest \ smallest key $\sim \Theta(n)$
remove the item $\sim \Theta(1)$

Ordered Array :

Insertion — find the proper position $\sim O(\log(n))$
add the item $\sim O(n)$

Better since there are never
more deletions than insertions

Ordered List :

Insertion — find the proper position $\sim O(n)$
add the item $\sim \Theta(1)$

Deletion — remove the first \ last item $\sim \Theta(1)$

Binary Search Tree :

Now you begin to know me 😊

Be careful
always dangerous.

a better option?



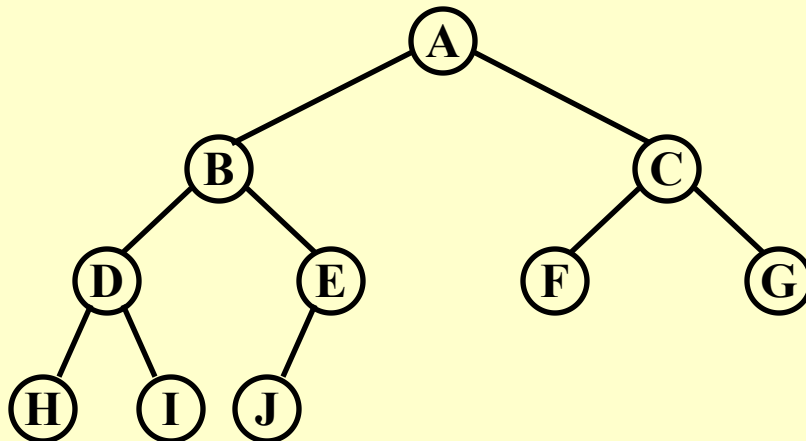
§ 3 Binary Heap

1. Structure Property:

【Definition】 A binary tree with n nodes and height h is **complete** iff its nodes correspond to the nodes numbered from 1 to n in the perfect binary tree of height h .

A complete binary tree of height h has between 2^h and $2^{h+1} - 1$ nodes. $\longrightarrow h = \lfloor \log N \rfloor$

❖ Array Representation : **BT** [$n + 1$] (**BT** [0] is not used)



BT

0	1	2	3	4	5	6
	A	B	C	D	E	F

7	8	9	10	11	12	13
G	H	I	J			

【Lemma】 If a complete binary tree with n nodes is represented sequentially, then for any node with index i , $1 \leq i \leq n$, we have:

$$(1) \text{ index of } \mathit{parent}(i) = \begin{cases} \lfloor i/2 \rfloor & \text{if } i \neq 1 \\ \text{None} & \text{if } i = 1 \end{cases}$$

$$(2) \text{ index of } \mathit{left_child}(i) = \begin{cases} 2i & \text{if } 2i \leq n \\ \text{None} & \text{if } 2i > n \end{cases}$$

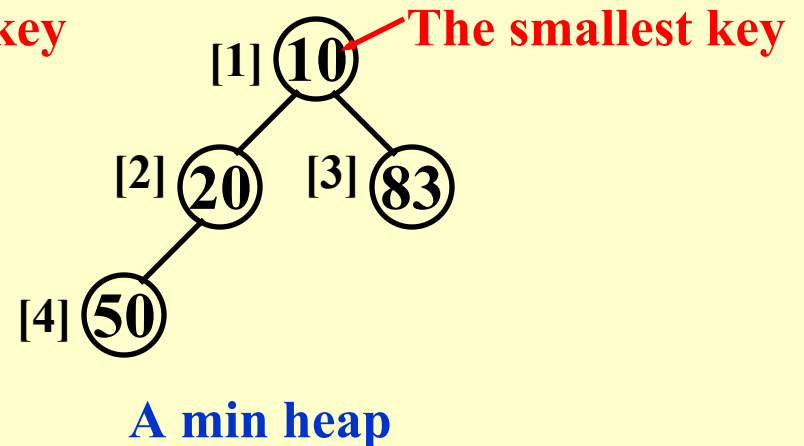
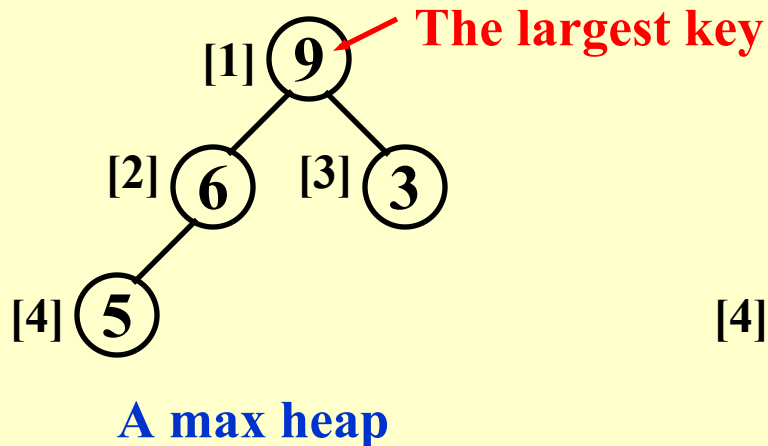
$$(3) \text{ index of } \mathit{right_child}(i) = \begin{cases} 2i + 1 & \text{if } 2i + 1 \leq n \\ \text{None} & \text{if } 2i + 1 > n \end{cases}$$

```
PriorityQueue Initialize( int MaxElements )
{
    PriorityQueue H;
    if ( MaxElements < MinPQSize )
        return Error( "Priority queue size is too small" );
    H = malloc( sizeof ( struct HeapStruct ) );
    if ( H ==NULL )
        return FatalError( "Out of space!!!" );
    /* Allocate the array plus one extra for sentinel */
    H->Elements = malloc(( MaxElements + 1 ) * sizeof( ElementType ));
    if ( H->Elements == NULL )
        return FatalError( "Out of space!!!" );
    H->Capacity = MaxElements;
    H->Size = 0;
    H->Elements[ 0 ] = MinData; /* set the sentinel */
    return H;
}
```

2. Heap Order Property:

【Definition】 A **min tree** is a tree in which the key value in each node is no larger than the key values in its children (if any). A **min heap** is a **complete** binary tree that is also a min tree.

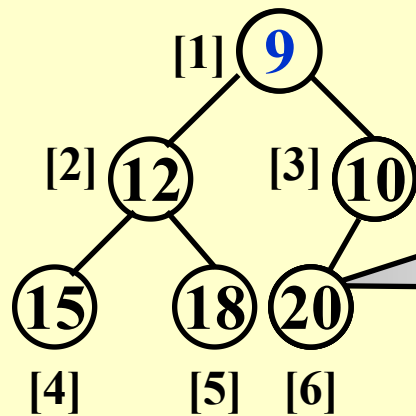
Note: Analogously, we can declare a **max** heap by changing the heap order property.



3. Basic Heap Operations:

👉 insertion

➤ Sketch of the idea:



The only possible position
for a new node
since a heap must be
a complete binary tree.

Case 1 : new_item = 21 $\textcircled{20} < \textcircled{21}$ ✓

Case 2 : new_item = 17 $\textcircled{20} > \textcircled{17}$ $\textcircled{10} < \textcircled{17}$ ✓

Case 3 : new_item = 9 $\textcircled{20} > \textcircled{9}$ $\textcircled{10} > \textcircled{9}$ ✓

/ H->Element[0] is a sentinel */*

void Insert(ElementType X, PriorityQueue H)

{

int i;

if (IsFull(H)) {

 Error("Priority queue is full");

return;

 }

for (i = ++H->Size; H->Elements[i / 2] > X; i /= 2)

 H->Elements[i] = H->Elements[i / 2];

 H->Elements[i] = X;

}

H->Element[0] is a *sentinel* that is no larger than the minimum element in the heap.

Faster than
swap

$T(N) = O(\log N)$

👉 DeleteMin

➤ Sketch of the idea:

Ah! That's simple --
we only have to delete
the root node ...

And re-arrange
the rest of the tree so that
it's still a min heap.

18
[4]



... to keep a
binary tree.

12

<

18

15

<

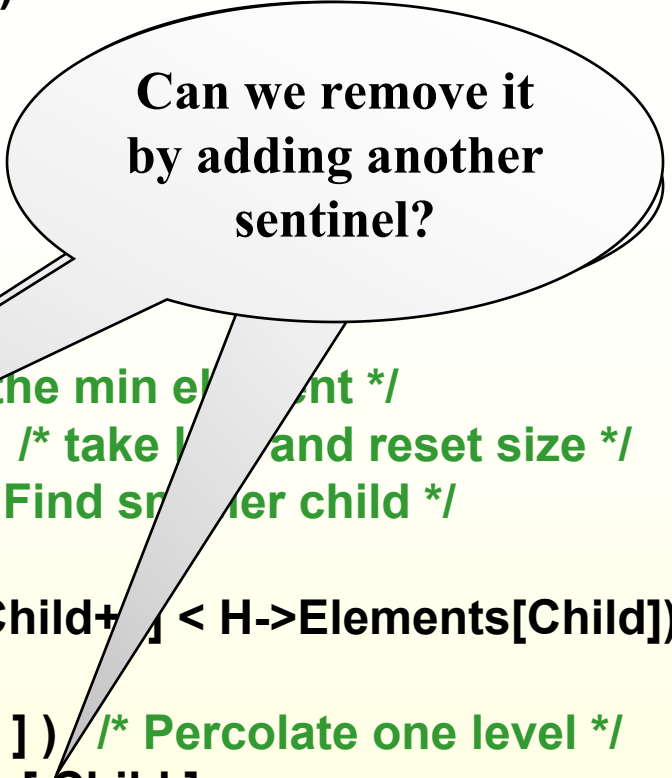
18

$$T(N) = O(\log N)$$

```

ElementType DeleteMin( PriorityQueue H )
{
    int i, Child;
    ElementType MinElement, LastElement;
    if ( IsEmpty( H ) ) {
        Error( "Priority queue is empty" );
        return H->Elements[ 0 ]; }
    MinElement = H->Elements[ 1 ]; /* the min element */
    LastElement = H->Elements[ H->Size-- ]; /* take last and reset size */
    for ( i = 1; i * 2 <= H->Size; i = Child ) { /* Find smaller child */
        Child = i * 2;
        if ( Child != H->Size && H->Elements[Child+1] < H->Elements[Child])
            Child++;
        if ( LastElement > H->Elements[ Child ] ) /* Percolate one level */
            H->Elements[ i ] = H->Elements[ Child ];
        else break; /* find the proper position */
    }
    H->Elements[ i ] = LastElement;
    return MinElement;
}

```



Can we remove it by adding another sentinel?

4. Other Heap Operations:

Note: Finding any key except the minimum one will have to take a linear scan through the entire heap.

☞ **DecreaseKey** (**P**, Δ , **H**)

Percolate up



Lower the value of the key in the heap **H** at position **P** by a positive amount of Δso my programs can run with highest priority ☺.

☞ **IncreaseKey** (**P**, Δ , **H**)

Percolate down



Increases the value of the key in the heap **H** at position **P** by a positive amount of Δdrop the priority of a process that is consuming excessive CPU time.

☞ Delete (**P**, **H**)

DecreaseKey(P, ∞ , H); DeleteMin(H)



Remove the node at position **P** from the heap **H** delete the process that is terminated (abnormally) by a user.

☞ BuildHeap (**H**)

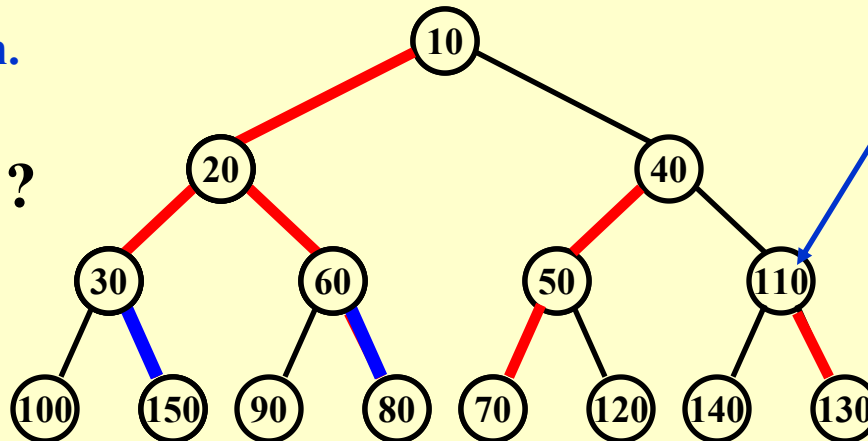
Nehhhhh that would be tooooo slow !



Place N input keys into an empty heap **H**.

150, 80, 40, 30, 10, 70, 110, 100, 20, 90, 60, 50, 120, 140, 130

$T(N) = ?$



PercolateDown (7)
PercolateDown (6)
PercolateDown (5)
PercolateDown (4)
PercolateDown (3)
PercolateDown (2)
PercolateDown (1)

【Theorem】 For the perfect binary tree of height h containing $2^{h+1} - 1$ nodes, the sum of the heights of the nodes is $2^{h+1} - 1 - (h + 1)$.

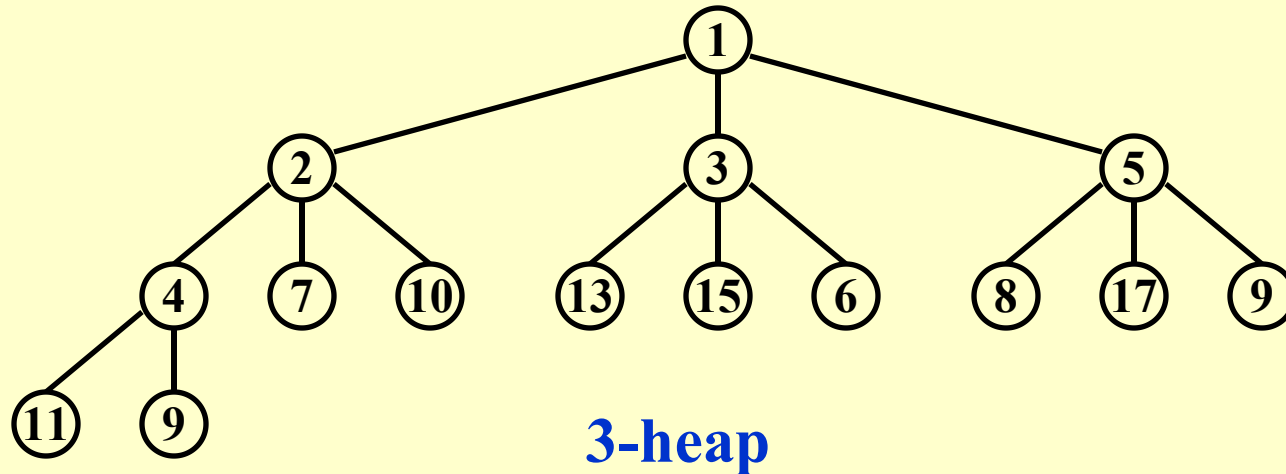
→ $T(N) = O(N)$

§ 4 Applications of Priority Queues

【Example】 Given a list of N elements and an integer k . Find the k th largest element.

How many methods can you think of to solve this problem? What are their complexities?

§ 5 d -Heaps ---- All nodes have d children



Question: Shall we make d as large as possible?

- Note:**
- ① DeleteMin will take $d - 1$ comparisons to find the smallest child. Hence the total time complexity would be $O(d \log_d N)$.
 - ② $*2$ or $/2$ is merely a bit shift, but $*d$ or $/d$ is not.
 - ③ When the priority queue is too large to fit entirely in main memory, a d -heap will become interesting.